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Innovative 3d Modeling and Well Optimization for Mauddud Formation Development in Greater Burgan Field

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Abstract

Traditional exploration strategies have primarily focused on conventional, well-understood geological features, i.e. simple anticlinal traps and thick reservoir formations, due to their favorable structural characteristics and economic viability.

However, contemporary challenges to increase production necessitate a paradigm shift towards less conventional exploration domains, emphasizing on geologically complex and previously underutilized regions that were historically deemed economically unfeasible. Development attempts of Mauddud formation in the Greater Burgan Area has been regarded as strategically favorable.

While appearing as a thin reservoir in Southeast Kuwait, one of the primary factors driving significant interest in Mauddud formation is its hydraulic connection with the underlying extensive main Burgan clastic reservoir, suggesting potential migration and mutual influence between the two formations.

KOC Reservoir Study Team (RST) and Innovation & Technology (I&T) in partnership with SLB have been tasked with the challenge of identifying optimal locations for the new well placement, defining suitable candidates for the re-entry side-tracks, and evaluating the feasibility of repurposing old wells - originally targeted for the lower formations – for production from the Mauddud Reservoir. A critical aspect of this initiative involves the integration of all available data to confirm formation connectivity and developing a DFN model.

This study presents an integrated 3D modeling approach that incorporates petrophysical, geophysical, geological, reservoir, and production data to optimize new well placement.

A new fine static geological model was built for the Greater Burgan Field with a primary focus on the Mauddud formation, while the Wara and Upper Burgan formations were included at a broader scale. Several novel rock classification approaches were implemented, and image log analysis facilitated the identification of structural features and vertical heterogeneity within the formation.

A supervised multivariate approach was employed to estimate the permeability model, allowing the integration of multi-domain datasets including wells logs and cores. Petrophysical rock typing, based on pore throat radius distribution, provided a detailed reservoir property characterization, improving well

candidate assessment and placement optimization. Furthermore, capillary pressure-derived water saturation estimates were utilized to model the original oil in place (OOIP).

At the conclusion of the comprehensive FDP study, two piloted horizontal wells were drilled and completed with multi-stage acid frac. The results of the DST were encouraging, validating the effectiveness of the revised FDP, and providing a foundation for the continued development of Mauddud formation, and paving the way for further additional wells in the future.

The successful Implementation of the integrated study on the Mauddud reservoir in the Greater Burgan field, despite numerous specific uncertainties, serves as a pioneering example for future advancements in the development of challenging, geologically complex formations.

Introduction and Background

The Mauddud Formation, a Late Albian carbonate member of the Wasia Group, constitutes a critical yet underdeveloped hydrocarbon reservoir in Kuwait's subsurface landscape. While the formation demonstrates favorable reservoir properties and prolific production in North Kuwait—where it appears as a thick, naturally fractured limestone with dual-porosity behavior—its expression in South and East Kuwait, particularly in the Greater Burgan area, poses considerable development challenges. In this region, the Mauddud formation thins to approximately 20-30 feet of net pay, displays high porosity (average ~18–20%), but suffers from low matrix permeability (typically <10 mD), creating a tight, heterogeneous reservoir system.

Historical attempts to unlock the potential of Mauddud-Burgan through horizontal well campaigns in the early 1990s met with limited success. The drilling programs encountered several operational and reservoir-related issues, including unexpected wellbore collapse, premature free gas production, poor fracture connectivity, and sub-commercial oil rates (W. H. Al-Bazzaz, et al. 2006). A key contributor to these issues was the structural complexity of the reservoir, characterized by thin, incompetent shale layers and sub-seismic faults, which resulted in well instability and hindered effective reservoir drainage. Additionally, the prevalence of a saturated pressure regime led to early gas liberation below bubble point, further disrupting oil flow and reducing productivity.

Despite these setbacks, recent advances in reservoir characterization, geomechanical modeling, and stimulation technologies offer a renewed opportunity to reassess and optimize the Mauddud-Burgan reservoir. The reservoir's complex stratigraphy—comprising high-energy dolomitized limestones (M1) and lower-energy glauconitic limestones (M2)—combined with indications of localized fracture systems and cavernous porosity, point to a need for more granular intra-reservoir zonation and high-resolution property modeling.

The renewed interest in the Mauddud-Burgan development reflects a strategic vision to elevate and sustain its production potential. This necessitates a comprehensive feasibility study and the implementation of cutting-edge technologies, data integration workflows, and optimized well designs that are tailored to the unique geological and mechanical behavior of the formation.

Reservoir and modeling challenges

The historical failure to effectively develop the Mauddud Formation in the Burgan area is rooted in a limited understanding of its geological heterogeneity and mechanical behavior. Early development programs were based on assumptions derived from North Kuwait analogues, overlooking the complex interplay of depositional facies, diagenetic overprint, and sub-seismic faulting that uniquely characterize the Burgan sector. Consequently, drilling plans underestimated the risks associated with structural instability, shale interference, and poor fracture continuity.

Moreover, prior modeling efforts lacked the resolution and zonation necessary to capture intra-reservoir variations in matrix permeability, porosity distribution, and natural fracture intensity. The inability to predict

sweet spots or optimize wellbore placement within mechanically stable and high-flow zones resulted in non-uniform production performance and abandonment of multiple drilled wells. Additionally, the emergence of early free gas, due to pressures below bubble point, complicated fluid handling, increased operational costs, and reduced economic feasibility.

Considering these limitations, a fit-for-purpose, multi-disciplinary approach is essential. This includes collecting and integrating static and dynamic data, refining zonation schemes, conducting geomechanical risk assessments, and implementing advanced stimulation designs. High-resolution reservoir modeling, combined with emerging technologies such as machine learning, image log analysis, and discrete fracture network (DFN) modeling, is required to improve reservoir understanding and optimize development strategies.

Ultimately, the problem is not one of resource scarcity, but of subsurface complexity and operational misalignment. This study aims to bridge that gap by delivering a technically and economically viable development framework for Maaddud-Burgan, informed by updated detailed geological studies and calibrated predictive models.

Methodology

The redevelopment of the Maaddud Formation in the Greater Burgan field was underpinned by a comprehensive suite of technical methodologies that addressed both geological complexity and reservoir heterogeneity. The study began with extensive data collection, quality control, and a gap analysis, leading to the construction of an enhanced static model incorporating nearly 3000 wells. This new model introduced zonation to distinguish the Upper and Lower Maaddud, a feature absent from the previous 2015 and 2019 models. In addition to the development of Ahmadi, Magwa, and Burgan areas, the Khashman zone was incorporated following a detailed review of ten wells (Fig. 1). However, uncertainty remains around the properties and free water level (FWL) in the Khashman area, warranting further investigation.

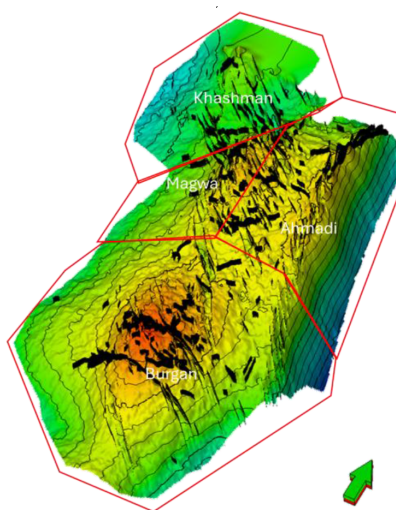


Figure 1—Top MA structural surface and interpreted faults.

The study was conducted with a broad scope that included:

- Geophysics
- Petrophysics
- Geology Static Modeling

- Borehole Geology
- Geomechanics 1D and 3D MEM
- Discrete Fracture Network Modeling
- Production Engineering
- Reservoir Engineering

Geophysics

A seismic cube that covers the Greater Burgan polygon takes its geographic data as input in the vertical direction from zero to 100 ft below the deepest point of the Shuaiba horizon. To make a complete structural framework model, a surface was added from zero depth including 6 seismic horizons interpreted without making any adjustments to the wells. Using a set of revised well tops the interpreted horizon of Mauddud was adjusted, quality checked and implemented into the framework. Interpreted faults were reduced from 663 to 489 and the fault framework modeling was finalized with 458 faults.

The primary objective of the seismic attribute analysis was to identify discontinuities associated with fault and fracture networks within the Mauddud interval. A suite of structural attributes was computed and integrated into a composite seismic volume—termed iFrac—which served as a proxy for fracture intensity. This volume was later calibrated (Fig. 2) against well fracture data and incorporated into the discrete fracture network (DFN) modeling workflow.

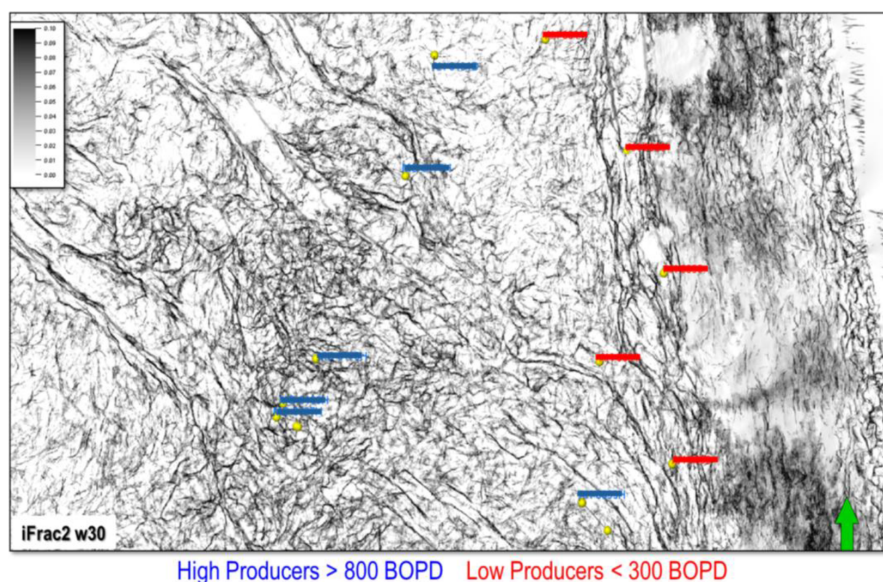


Figure 2—A generic consistent curvature workflow for mapping regional discontinuities or fracture corridors and Map view of the resulting Seismic attribute iFrac

The analysis commenced with structural smoothing of the pre-stack depth migrated (PSDM) seismic data to enhance spatial continuity and suppress residual noise. Structural smoothing parameters were carefully calibrated to avoid over-filtering and preserve coherent signals. This conditioning step ensured optimal input quality for subsequent structural attribute calculations.

The next step involved computation of Consistent Dip, a volumetric dip estimation technique based on iterative global optimization. Compared to conventional instantaneous dip methods, consistent dip improves lateral and vertical resolution and enhances the reliability of structure-oriented attributes. The resulting dip volumes served as the foundation for multiple attribute computations, including curvature, residual dip, and RMS dip.

A range of seismic attributes was computed to highlight discontinuities indicative of faults and possible fracture corridors:

- Residual Dip and RMS Dip: These attributes emphasize local orientation changes and were input into an ant tracking workflow using aggressive settings to delineate potential fracture swarms.
- Curvature (K1, K2): Derived from inline and crossline dip volumes, curvature attributes quantify the bending of seismic reflectors. The K1 (maximum curvature) volume was selected for its ability to highlight fold-related fracture zones and entered into the ant tracking workflow.
- Variance and Chaos: These classical discontinuity attributes were generated using different parameter sets. Variance accentuates amplitude variability along vertical structures, while chaos identifies discontinuity in all directions. Both were useful in highlighting larger-scale fault zones.

Each attribute volume was normalized using the following equation:

$$\text{SeismicNormalized} = \frac{\text{SeismicInput} - \mu}{\sigma}$$

where μ and σ are the mean and standard deviation of the attribute volume, respectively.

The final iFrac volume was generated through Petrel's volume calculator by averaging 4 normalized seismic attribute volumes that exhibited the strongest discontinuity signatures (suggesting that the contribution of Chaos may introduce localized noise rather than improving fracture resolution it was excluded):

$$\text{iFrac} = \frac{\text{ResidualDipAT} + \text{RMSDipAT} + \text{CurvatureK1AT} + \text{VarianceAT}}{4}$$

Using the final structural model adjusted with the well tops, the fracture well data (from BHI interpretation) were compared with the image of the seismic attribute (Residual Dip) at Fig. 3.

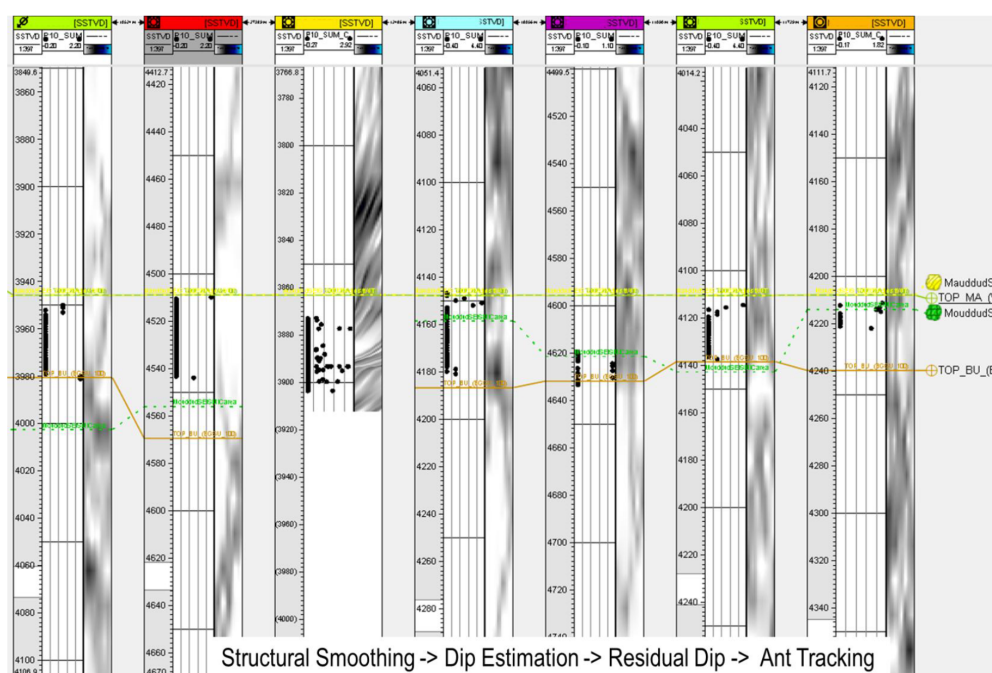


Figure 3—Comparison between wells with the fracture intensity well log and the Ant Tracking attribute of seismic discontinuity Residual Dip.

Good correlation is observed between fracture well data and the generated seismic attribute (Residual Dip) in most of all the wells.

Petrophysics

Reservoir rock types (RRTs) were redefined using pore throat radius data and Mercury Injection Capillary Pressure (MICP), reducing the classification from six types in the previous study (2020 B.Vincent et al.) to four (Fig. 5). This refinement contributed to a more robust static-to-dynamic property transfer.

The available rock-typing analysis was updated with new core data, considering routine core analysis data of 39 wells and the new estimated permeability from well log data. Net confining stress of 2300 psi is usually used for the entire section of Wara-Maaddud-Burgan, and in this study area, core porosity was comparable with log total porosity.

The pore throat size distribution explains the low reservoir quality. Mercury injection pore size distribution plot, as shown in Fig. 4, allows to corroborate that the Maaddud formation is dominated by microporosity pore throats as well as mesoporosity pore throats. A petrophysical rock typing (PRT) using the Winland 35 method was developed. This technique, based on pore throat radius (PTR), is closely associated with rock's petrophysical properties such as porosity, permeability, and irreducible water saturation. The new petrophysical rock typing was applied at core/log level and compared with the existing facies interpretation.

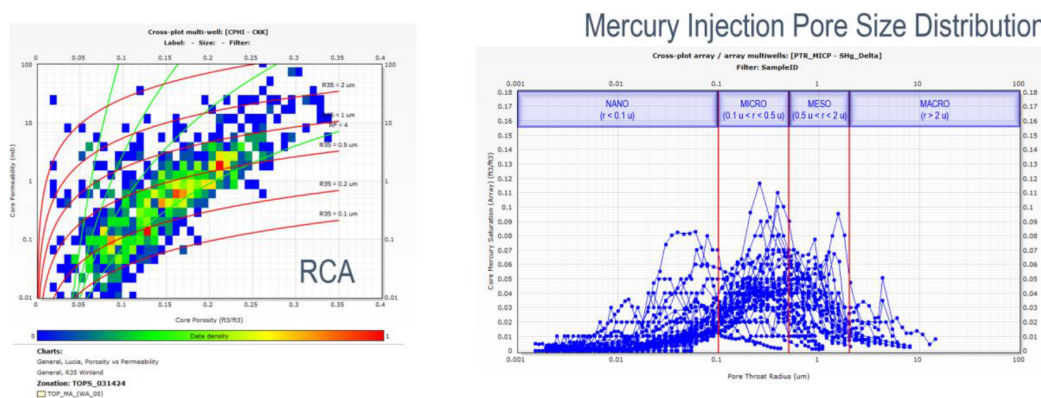


Figure 4—Mercury Injection Pore Size Distribution

As illustrated in Fig. 5, there is a distinct petrophysical correlation between the previous studies facies and the proposed petrophysical rock typing.

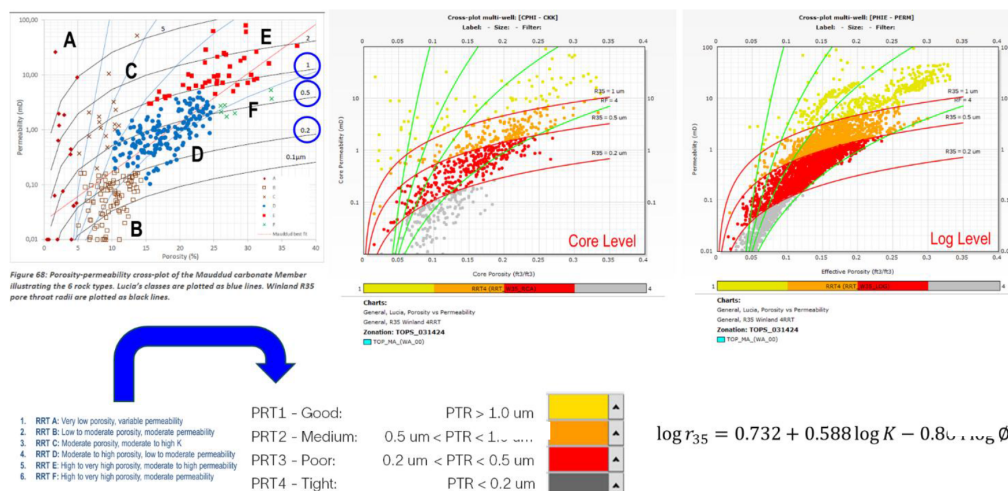


Figure 5—Update Rock Typing for Maaddud Reservoir

Geological Modeling and Static Model Construction

In constructing the static geological model for this reservoir, data from 2943 wells across the field were integrated for correlation and property analysis. Two key structural surfaces (top of the Mauddud and Wara formations) were interpreted along with fault polygons to delineate the zone of interest in the model. Intra structure, Top of the Burgan Upper Sand and the bottom of the model were calculated based on the Top of the Mauddud structure. Based on variations in properties, the Mauddud Formation was internally subdivided into sub-zones (i.e. upper and lower Mauddud by Intra structure) to capture heterogeneity. The bounding surfaces with the overlying Wara and underlying Burgan formations were adjusted in the model according to reservoir simulation layering requirements. These adjustments ensured the Mauddud zone tied in properly with the Wara and Burgan zones as required by the dynamic simulation grid. The Wara and Burgan formations themselves were included in the model for context; their rock typing and property modeling parameters were adopted from a previous 2015 static model to maintain consistency with earlier studies and field development plans.

Reservoir rock types (RRTs) were populated in the 3D static model to honor the facies and petrophysical heterogeneity of the Mauddud reservoir. [Fig. 6](#)

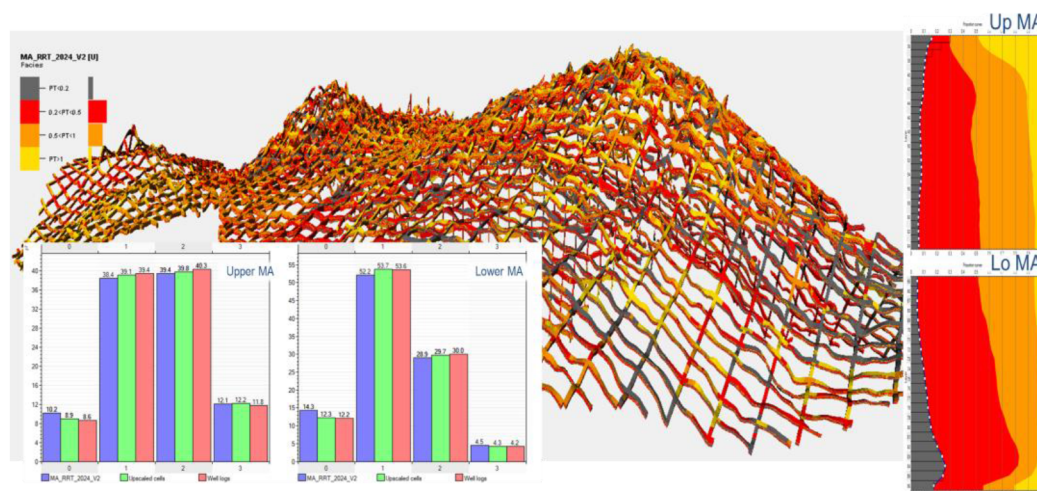


Figure 6—3D distribution of RRTs (property), vertical proportion curves and histograms of RRTs (property- upscaled- well logs)

The RRT model was used to constrain petrophysical property simulation. Further conditioned to porosity: zone-specific porosity–permeability regression relationships (derived from core data) were used to guide the base "matrix permeability" in each zone. In practice, this means the average permeability in a cell was first estimated from the cell's porosity via a regression formula (per zone or per rock type as at [Fig. 7](#)). Cutoff values obtained from the rock typing process (e.g. porosity thresholds between rock types, as noted above) were used to truncate or limit the property ranges for each rock type, preventing the simulation from generating values outside realistic bounds.

Permeability was populated in the 3D grid using stochastic simulation approach. A sequential Gaussian simulation (SGS) algorithm was employed to generate multiple realizations of the permeability field. The simulation was controlled by the rock types – meaning that each rock type was assigned its own statistical distribution and variogram for permeability, so the final model honors the distinct permeability ranges of each RRT.

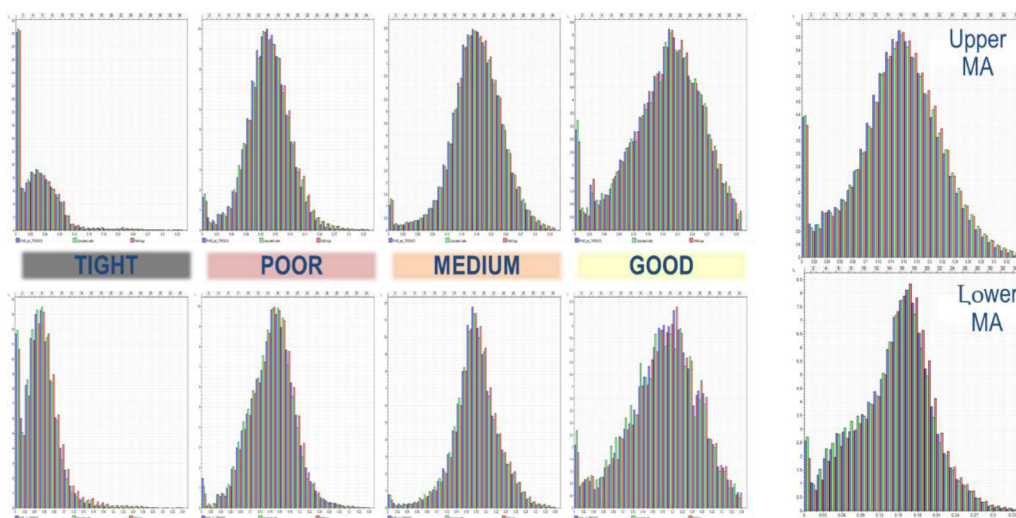


Figure 7—Porosity distribution by RRT in Upper MA and Lower MA and together combined by zone

Water saturation (S_w) was modeled using a saturation-height function based on the Brooks and Corey capillary pressure correlation. In this approach, the irreducible water saturation (S_{wirr}), capillary entry (displacement) pressure (P_e), and pore-size distribution index (λ) for each rock type or zone were defined according to the Brooks–Corey model parameters through a workflow to calculate water saturation as a function of height above the free water level. A unique free water level (FWL) was assumed for the entire Greater Burgan field's Mauddud reservoir in the model, based on field pressure data.

With the static model fully populated (structure, rock types, porosity, permeability, and saturations), volumetric calculations were carried out to estimate original oil in place (OOIP) with the current FWL assumption. The significant volume, contained in a relatively thin reservoir interval, underscores the importance of accurate geological modeling and property distribution in the Mauddud for forecasting and development planning.

Geomechanics

Geomechanical characterization played a critical role in the development strategy. One-dimensional Mechanical Earth Models (1D MEMs) calibrated with static and dynamic mechanical data such as closure pressure, Young's Modulus (YME), Poisson's Ratio (PR), uniaxial compressive strength (UCS), tensile strength (TSTR) and friction angle modelling. (M. Haas et al. 2024) These were used to generate a three-dimensional MEM to estimate the in-situ stress field in structural equilibrium. It incorporated a DFN based on fracture prediction techniques, with faults and fractures embedded into a 25 million grid cell geomechanical grid. These results were derived by a mix of co-kriged properties, machine learning from extensive laboratory and field data. Property modelling in the 3D MEM was performed using the Machine Learning embedded algorithm and guided using seismic data (interval velocity and seismic inversion data) as well as tertiary data such as laboratory tests, k-layering and the upscaled 1D MEM logs.

Blind test models cross-validated against field data proved the model calibration, though friction angle (FANG) correlation was limited by data availability. The vertical stress gradient averages at ~ 0.98 psi/ft; pore pressure ranges from 0.44–0.48 psi/ft. The maximum horizontal stress is oriented ~ 40 – 50° from North.

To better capture the role of natural fractures, a high-resolution Discrete Fracture Network (DFN) was developed integrating geomechanical stress inversion, fault characterization, borehole image analysis, and seismic-derived attributes. The goal was to capture the spatial heterogeneity of natural fractures and improve representation of fracture-related flow properties across the field.

Fracture modeling was driven by two primary mechanisms:

1. Fault Natural Fracture Prediction (Fault-NFP) based on paleo-tectonic stress inversion, and
2. Fault Damage Zone (FDZ) intensity modeling using empirical decay laws.

An additional fracture set was incorporated from filtered seismic attribute patches, contributing to the deterministic fracture population.

The Fault-NFP approach involved simulating stress perturbations around seismically mapped faults, constrained by borehole image and seismic-derived fracture orientations. Paleo-stress conditions were estimated using iBem3D through multi-scenario stress inversion, revealing two dominant tectonic episodes. These episodes generated four main fracture sets—shear and tensile joints in NE-SW and NW-SE directions—consistent with the region's Alpine tectonic history. Fig. 8

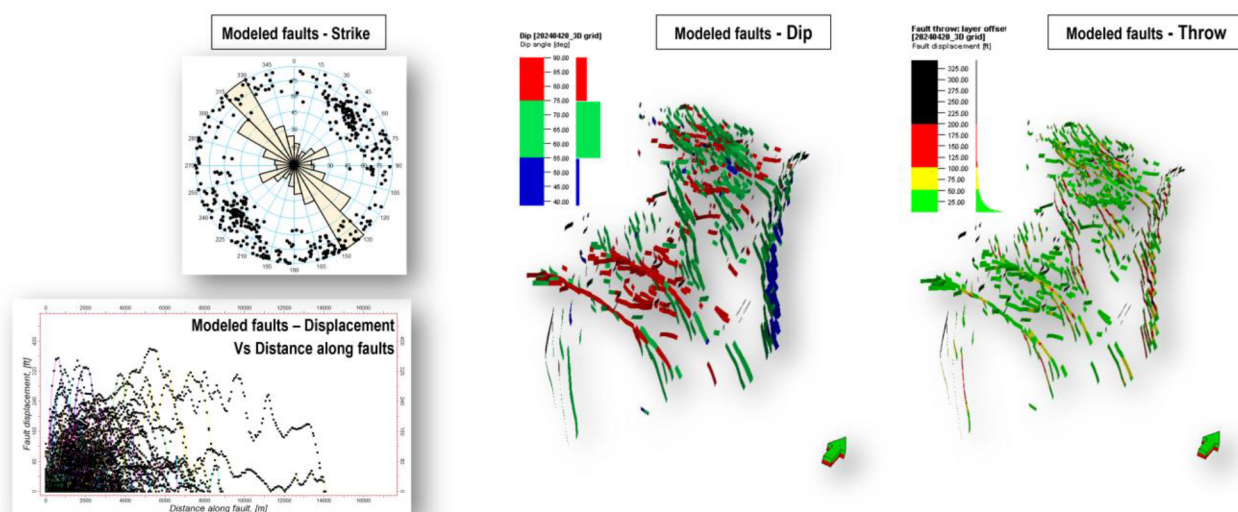


Figure 8—Fault analysis for MA reservoir

The FDZ driver modeled fracture intensity decay as a function of distance to fault, with variable exponent and scaling constants reflecting heterogeneous fault displacement. Fracture orientations within damage zones were determined using local fault geometry, while the width of influence was defined at ~100 m on either side of fault traces.

Six fracture sets were ultimately modeled: four from NFP, one from FDZ, and one from seismic patches. Fracture apertures were distributed stochastically using statistical parameters derived from borehole image data. Two DFN scenarios were generated—one with constant fracture concentration and extended lengths, the other with well-derived concentration and reduced length distributions. Both models were upscaled using an Oda-corrected method to derive fracture porosity, permeability tensor (σ), and directional permeability (K_i).

Model validation against borehole images and 3D formation scanning (3DFFS) confirmed a strong match in fracture orientation and intensity, with dominant NE-SW, NW-SE, and N-S trends. A clear spatial distinction was observed between Magwa/Burgan (NE-SW/E-W) and Ahmadi (N-S/NW-SE) regions. Fractures were predominantly high-angle ($>30^\circ$) and fault-related, supporting their tectonic origin.

This integrated fracture modeling approach—leveraging structural, mechanical, and seismic drivers—demonstrates the value of multi-driver DFN construction in capturing fracture complexity (Fig. 9). The resulting models offer a robust framework for assessing fracture connectivity and guiding dynamic simulation, with further optimization recommended for fracture length calibration to enhance permeability representation.

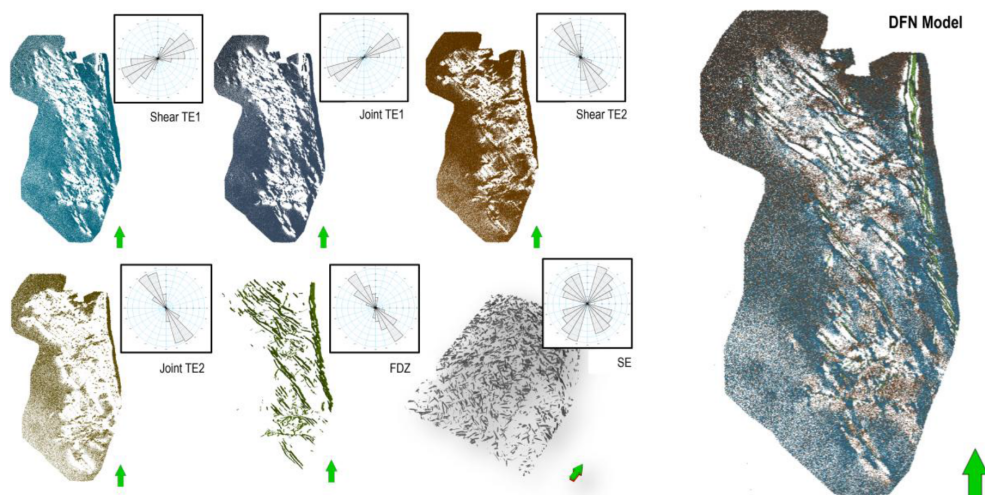


Figure 9—Fracture sets in DFN fracture model

Reservoir Engineering

Following dynamic simulation and calibration, a sweet spot property map was derived by integrating five key parameters: reservoir pressure (Pres), minimum horizontal stress ($1 - Sh_{min}$), fracture permeability (k_{frac}), reservoir transmissibility (kh), and oil saturation ($Soil$). These were combined using a weighted equation:

$$\text{Sweet_Spot_Property} = 0.2\text{Pres} + 0.2(1 - Sh_{min}) + 0.2k_{frac} + 0.2kh + 0.2 * Soil$$

The resulting sweet spot distribution provided a spatial reference for optimal well placement (Fig. 10 and Fig. 11).

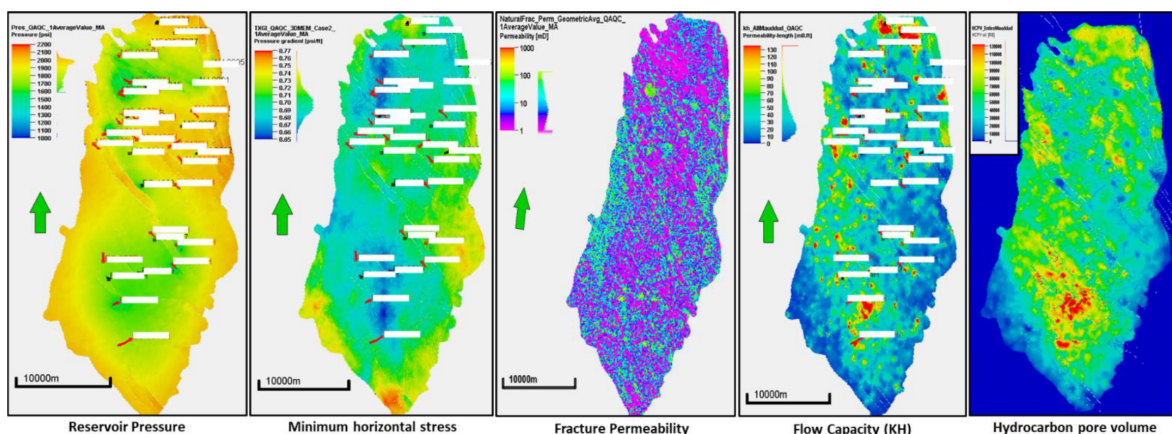


Figure 10—Maps of Average Reservoir Pressure, Minimum Horizontal Stress, Fracture Perm, Matrix KH and HCPV in Maaddud

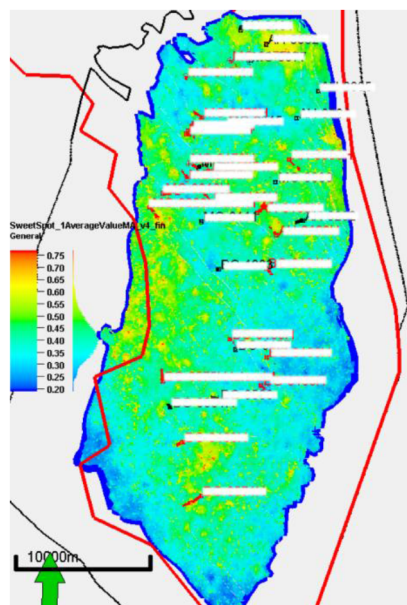


Figure 11—Sweet spot map Mauddud Reservoir

In parallel, a short-term production enhancement campaign was initiated by reviewing over 100 wells using nodal and rate transient analysis.

The combination of data-driven reservoir modeling, machine learning applications, and fracture-focused simulation not only enhanced the accuracy of production forecasting but also provided a basis for optimized horizontal well placement and stimulation design. Moreover, the identification of short-term production enhancement opportunities demonstrates the operational value of this integrated methodology beyond long-term development planning.

With the technical foundation established, the following section presents the key findings, production forecasts, and field development strategies derived from the applied workflow—highlighting both near-term gains and long-term production potential across the Mauddud reservoir.

Results and Innovations

The integrated study of the Mauddud Formation in the Greater Burgan area yielded significant technical insights and introduced multiple innovations in reservoir characterization, geomechanics, and development planning. These findings form a robust foundation for integrating the static geological model into dynamic reservoir simulation and comprehensive field development planning. It emphasizes key considerations in the modeling of thin, laterally and vertically heterogeneous reservoirs, where resolution of internal architecture, facies continuity, and layering is critical. The workflow also ensures the accurate incorporation of overlying and hydraulically connected underlying formations, allowing for a more realistic representation of vertical communication pathways and potential flow barriers within the broader reservoir system.

The Mauddud static reservoir model was refined through detailed intra-reservoir zonation, which sharpened the spatial distribution of petrophysical properties and fed directly into a field-scale 3D Mechanical Earth Model (MEM) calibrated with drilling events and caliper data. High-resolution porosity–permeability transforms, MICP, and well-testing show that, although Mauddud possesses substantial storage porosity, its matrix permeability alone cannot sustain production, a finding corroborated by core and image logs that reveal a dual-porosity system punctuated by cavernous pores in the upper intervals. Integrating the MEM with a Discrete Fracture Network informed by fault-damage zones, natural-fracture prediction, and seismic-extracted sets captured six fracture families with stochastically distributed apertures, providing a realistic basis for flow simulation. Geochemical fingerprinting, MDT gradients, and production performance

confirm hydraulic communication with the underlying BGSU—and, in roughly one-fifth of completions, water inflow from BGSU and/or Wara—indicating that future development must adopt a multi-reservoir perspective. Simulated horizontal-well scenarios with matrix acidization outline optimal well spacing and emphasize that limited fracture–matrix interchange demands targeted stimulation to prolong field life, while pressure depletion and solution-gas drive remain the dominant recovery mechanisms. Sweet-spot mapping of fracture corridors and pressure-sink avoidance has already defined two pilot wells, demonstrating that a technology-integrated workflow can unlock the full potential of the technically challenging yet strategically vital Mauddud–Burgan carbonate system in Kuwait.

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